SEPM-001 LAB REPORT

Comparison of Thread Counting Using Static and Non-Static Variables

Student name: _____

Student ID / Roll: _____

Course: Software Engineering and Project Management

Date: _____

GitHub repository URL: _____

GitHub directory path: Lab_Thread_Counter

1. Objective

[Write the objective in your own words.]

2. Tools and Environment

Java/JDK version: _____ Operating system: _____

IDE or terminal used: _____

3. Handwritten Source Code

[Insert clear scanned pages or photographs of your handwritten Java source code in the final report.]

4. Test Inputs

Test	Threads N	Increments/thread K	Expected N × K
TC1	1	1,000	1,000
TC2	2	10,000	20,000
TC3	5	10,000	50,000
TC4	10	50,000	500,000
TC5	20	50,000	1,000,000
TC6	50	50,000	2,500,000
TC7	100	50,000	5,000,000

5. Program Outputs

[Insert screenshots of your actual thread-safe and unsafe program runs. For the unsafe mode, include at least five runs for every test case. Do not fabricate or alter outputs.]

6. Result Analysis

Record actual results in the accompanying results.csv file or insert your completed result table here.

Expected count: $E = N \times K$

Absolute difference: $D = |S - I|$

Percentage difference: $P = (|S - I| / I) \times 100$, when $I \neq 0$

7. Answers to Analysis Questions

1. What is the difference between a static variable and a non-static variable in Java?

_____

2. Why do all threads share the same static counter?

_____

3. Why does each thread have its own non-static counter in this experiment?

_____

4. Why is join() required before calculating the final counts?

_____

5. Why can the unsynchronized static count be lower than the expected count?

_____

6. Does increasing the number of threads always increase the percentage difference? Explain using your observations.

_____

7. Why might two runs with the same number of threads produce different results?

_____

8. What changes when AtomicLong is used instead of a regular long?

_____

9. How would you modify the program so that all threads share one instance counter as well as the static counter?

_____

8. Observation

[Describe what your recorded results show. Explain any variation between repeated unsafe runs.]

9. Conclusion

[Write a conclusion supported by your actual outputs. Thread-safe totals should match; unsafe lost updates are possible but not guaranteed in every run.]